3.19 POWER MECHANICS (447)

Power Mechanics Paper

SECTION A marks

*Answer **all** the questions in this section*

a List **three** factors to be considered when putting up a motor vehicle spare parts shop. marks

b Explain **two** reasons why it is important to study power mechanics. marks

a State the full terms represented by the following engineering drawing abbreviations:

i CL

ii Ø

iii CSK

iv AF

marks

b Name **two** classes of fire and for each class, identify **one** appropriate commercial fire extinguisher. marks

a State **two** advantages of self tapping screws over ordinary screws. marks

b i Sketch an adjustable spanner. mark

ii State where long nose pliers may be used in a small engine. mark

a Explain **one** purpose of each of the following energy convertors in a motor vehicle:

i alternator mark

ii photo voltaic cells mark

b State **two** effects of adding each of the following alloying materials to carbon steel:

i Nickel mark

ii Molybdenum mark

With the aid of sketches, differentiate between a cylinder in line and a V cylinder engine block. marks

Figure 1 shows a sectional view of a Wankel engine. Describe **one** cycle of its operation with reference to C and D. marks

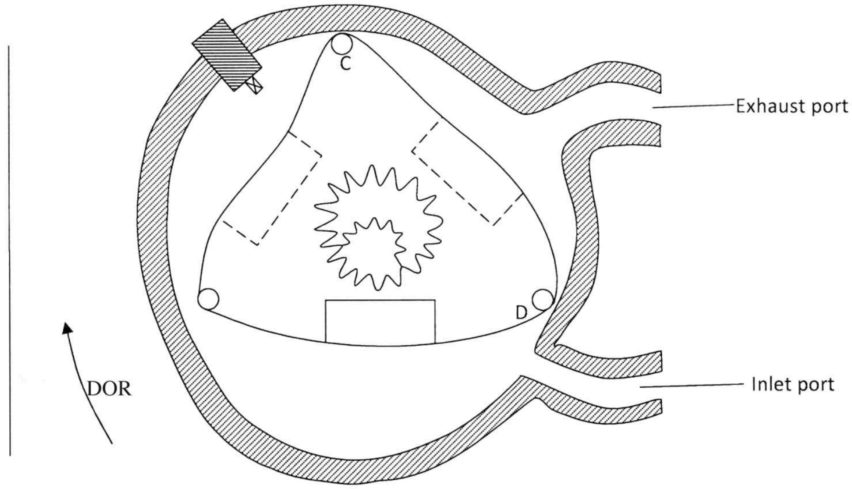

Figure 1

a Name the main components of the power transmission system of a motor vehicle. marks

b Explain the reason why modern vehicles are designed with collapsible steering columns. marks

(a) Briefly explain the process of hard soldering. (3 marks)

b Explain the following terms as used in drum brake operation

i leading shoe

ii trailing shoe marks

a State the purpose of the ply rating of a tyre. marks

b State **two** advantages of an independent suspension system over rigid beam suspension system. mark

Sketch a dipped beam light path having an offset filament and label its parts. (3 marks)

SECTION B □□□□ marks□

Answer question □□ and any other ***three*** *questions□*

□□□ Figure □□ shows an isometric view of a Vee block resting on one side□

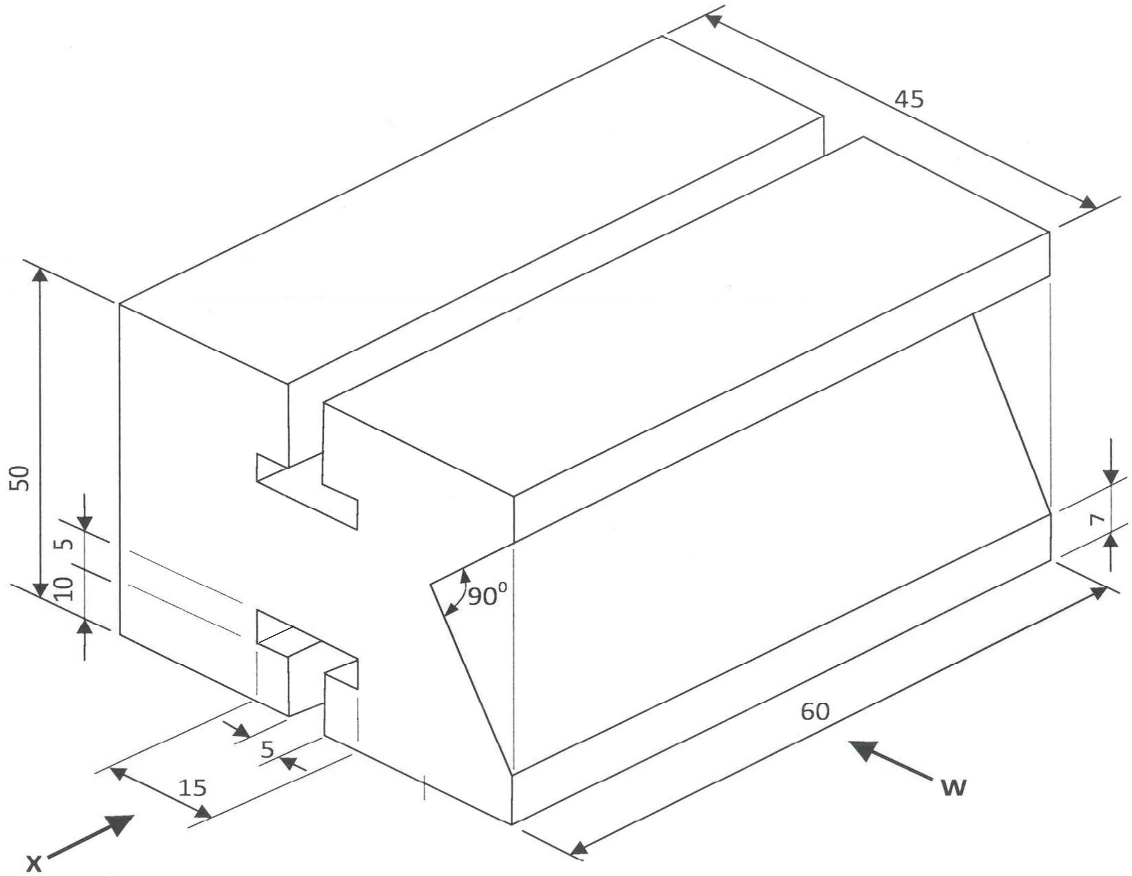

Figure 2

Draw full size, in first angle projection, the following views:

□a□□ front elevation in the direction of arrow **W**□

□b□□ end elevation in the direction of arrow **X**□

□c□□ Plan□

□Use A□ paper provided□□ □□□ marks□

Figure shows a component of the power transmission system of a motor vehicle.

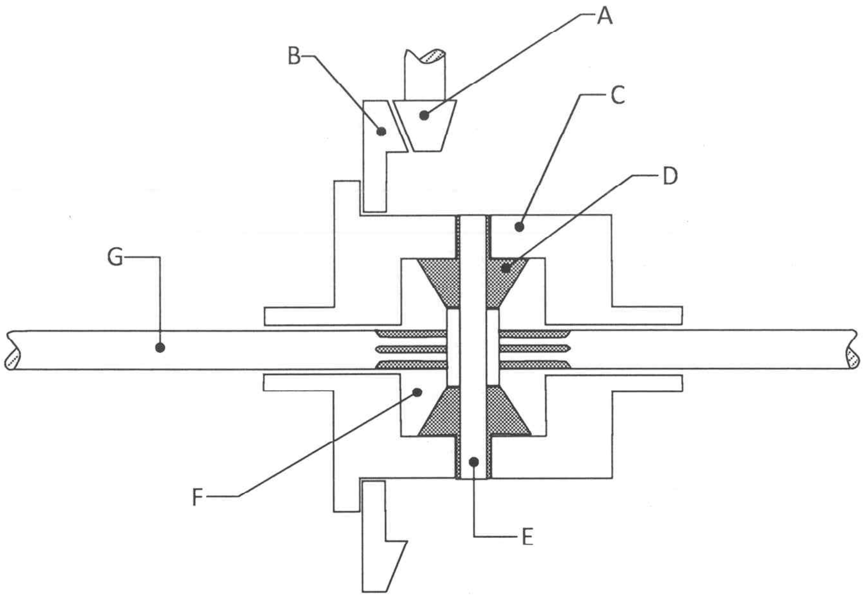

Figure 3

(a) Name the component. ($\frac{1}{2}$ mark)

(b) Name parts labelled **A** to **G**. ($\frac{1}{2}$ marks)

(c) Explain how the component operates. (marks)

With the aid of a labelled diagram, explain the operation of an overhead valve engine train whose camshaft is in the engine block. (marks)

With the aid of labelled diagrams, explain the operation of a four stroke compression ignition system. (marks)

(a) State **three** advantages of disc brakes over drum brakes. (marks)

(b) Sketch a sectional diagram of a disc brake assembly and label six parts. (marks